Track 45: CATIA Mechanical Design

June 2026

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Tier 3 — Specialized | Departments: Mechanical · Mechatronics · Aerospace elective

> ***Skill path:*** *Part Design → Assembly → Surfacing → Sheet Metal → Kinematics → Drafting → DMU Review*

Related: Track 52 — SolidWorks · Track 46 — ANSYS

Track Overview

Who this is for: Mechanical/Mechatronics students targeting automotive, aerospace, and high-end product design roles where CATIA is industry standard.

Prerequisites: Engineering drawing; Track 52 SolidWorks or equivalent CAD experience helpful.

Tools: CATIA V5/V6, ENOVIA intro, rendering workbench.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	CATIA awareness	Part + assembly + drawing
1 Month	80–100	Junior CATIA designer	Parametric assembly project
3 Months	250–300	Internship-ready	Surfacing + kinematics capstone
6 Months	500–600	Design engineer portfolio	Automotive/aerospace-style project

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Sketcher, pad/pocket, dress-up features	5 part models
D3–4	Assembly constraints, sub-assemblies	Mechanism assembly
D5–6	Drafting, GD&T, title blocks	Drawing package
D7–8	Surface design intro, hybrid modeling	Surfaced component
D9–10	Capstone	**CATIA design project**

1-Month Foundation

Week	Topics	Project
W1	Advanced part design, parameters, formulas	Parametric parts library
W2	Assembly design, clash detection	Multi-component assembly
W3	Sheet metal, weld design intro	Fabrication model
W4	DMU kinematics, motion simulation	**Kinematic mechanism project**

3–6 Month Progression

M1–M4: Advanced surfacing (Class A intro) + large assembly management + industry case study + portfolio (6-month).

Assessment Rubric

Weekly Quiz 20% | CAD labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Tier 3 **CATIA** brochure course. Pair with Track 46 ANSYS for simulation validation.